Vertical turbines more effecient

Research has found that vertical turbines are 15% more efficient than traditional turbines.

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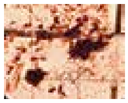

Fishes swallowing plastics

Scientists have found that fish have been swallowing microplastics since the 1950s.

http://dgit.in/may21-30

system. Scientists knew how large the nucleus was, but they had many other questions, what is it made of? Is it similar or different to comets in our own solar system? What can we find out about this alien star that forged the comet?

The ALMA and Hubble observations together pointed at something very odd, at least in comparison to local comets. Borisov was outgassing carbon monoxide in quantities not seen by Solar System comets. The carbon monoxide to water outgassing ratio was between 35 to 155 percent, which is a lot considering the average value in Solar System planets is 4 percent. The comets out in the Oort cloud though have ratios in the range of 10 to 24 percent. In fact, Borisov is one of the most carbon monoxide rich comets known, with the only one richer in carbon monoxide being an oddball comet called "the blue comet" or C/2016 R2 (PanSTARRS), with a really strange chemical composition, additionally being rich in nitrogen and poor in dust.

Carbon monoxide is much more volatile than say water, and very easily converts to the gaseous form at very low temperatures. At about a 100 AU, or more than twice the distance Pluto is from the Sun, most of the carbon monoxide escapes. Water remains ice till about 1.9 AU, where the asteroid belt is located. There are no comets with such high amounts of carbon monoxide outgassing so close to the Sun, as Borisov was observed during its perihelion. The observed outgassing could only occur if the carbon monoxide locked deep within the nucleus of the comet was trapped within shells of ice, that melted as the comet moved closer to the Sun, allowing the outgassing of the material trapped from the formation of the origin system of the comet. The Hubble and ALMA observatories were looking at ancient material from the birth of an alien star system.

These findings allowed the scientists to figure out that the comet was probably born in the outer reaches of the system around a red dwarf star, in the accretion disc as the star and the system were forming. The comet would have been formed in a region rich in carbon monoxide and icy bodies. It might even be the remnant of a planet that had a lot of carbon monoxide on the surface, that was subsequently pulverised by a colossal collision in the outer disc of the star. The location of the object in the outer disc of the star is for two reasons, the first being how cold the conditions needed to have been for the carbon monoxide to be trapped, and the second is that the object itself was ejected out of the system and thrown into an interstellar pinball machine, jumping between systems across the galaxy. The cool red dwarf stars radiate in the temperature range and luminosity range that is suitable for an object such as Borisov to form. These are fainter, less massive and smaller stars than the Sun, and hence form more compact systems. The HST team in their studies pointed out that the dynamical models of the Solar System indicate that there are comets in the Kuiper Belt that are captured objects from other star systems, which are the comets with the higher carbon monoxide content.

Researchers from ESO used the Very Large Telescope (VLT) to study Borisov at its close approach, and what that team found was that the material pouring out of the comet was ancient, like 4.5 billion years ago ancient, and the only other similar comet from the very origin of the Solar System that we know of was Hale-Bopp. The conclusion was derived from very precise characterisation of the polarisation of the light from the object. In fact, Borisov was even more pristine than Hale-Bopp, which means that the outgassing was made up of the same matter that the comet was formed in, unaltered by any further processes. The studies showed that even though the two comets came from very different star systems, the conditions in which they formed were very similar. The further implication is that this interstellar visitor, was created in the conditions that were very similar back when our Solar System was just born.

The two known interstellar visitors, 'Oumuamua and Borisov have together ushered in a new era of exoplanet studies, one that involves observing directly the debris of other star systems. All these studies together have primed scientists for the next interstellar visitor to come our way, and in Arthur C Clarke's Rendezvous with Rama, these always come in threes.

Credit: HST/NASA

Two photos of Borisov taken by Hubble, with a galaxy in the background on the left, and at its closest approach to the Sun on the right.